

## KNN.

Training Dataset  $\rightarrow m$  samples

new test sample  $P$

- ① choose a value  $K$
- ② find  $K$  most nearest samples of  $P$  in dataset.
- ③ Regression problem: mean of labels of  $K$  nearest samples.  
classification problem: mode (majority voting) of labels of  $K$  nearest samples.

$$P(y=1|x) = \frac{\text{no. of neighbors with } y=1}{K}$$

if  $K=5$ , 3 neighbors with  $y=1$

$$P(y=1|x) = \frac{3}{5} = 0.6$$

Regression.  $K$  NN

$$\hat{y} = \frac{1}{K} \sum_{i=1}^K y_i$$

- ① it can create non-linear boundaries.
- ② solve both regression and classification (binary or multiclass).

In Linear/Logistic regression, we save weights  $w_0$

In KNN:

- ① Training data
- ②  $K$
- ③ preprocessing.

$m$  samples,  $d$  dimension:

$$O(m \cdot d)$$

## \* Distances:

① Euclidean:

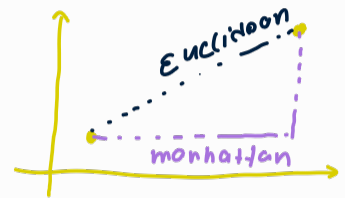
$$d(x, z) = \sqrt{\sum_{j=1}^p (x_j - z_j)^2}$$

$$x = (2, 3), z = (5, 7)$$

$$= \sqrt{(5-2)^2 + (7-3)^2}$$

$$= \sqrt{9 + 16}$$

$$= \sqrt{25} = 5$$



② Manhattan distance:

$$d(x, z) = \sum_{j=1}^p |x_j - z_j|$$

$$d = 7$$

③ Minkowski distance:

$$d(x, z) = \left( \sum_{j=1}^p |x_j - z_j|^q \right)^{1/q}$$

④ Cosine distance:

similarity:

$$\cos(\theta) = \frac{x \cdot z}{\|x\| \|z\|}$$

distance:  $1 - \cos(\theta)$

Scaling is important:

X input

$$X_{\text{norm.}} = \frac{X - \mu}{\sigma}$$

min-max norm.

$$X_{\text{norm}} = \frac{X - X_{\min}}{X_{\max} - X_{\min}}$$

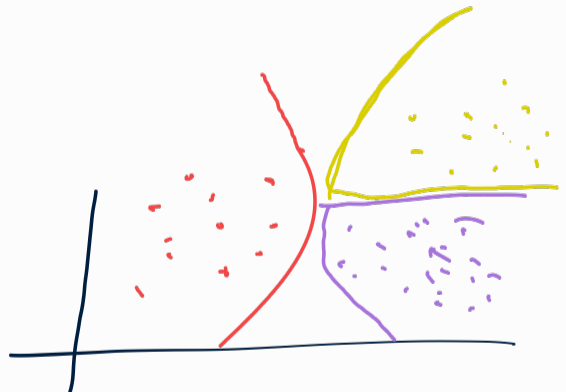
flow for KNN.

- ① split the data
- ② fit scaler on train data.
- ③ Transform the validation/test data based on scaler.
- ④ choose  $k$ , distance metric.
- ⑤ fit KNN.

\* Choosing  $k$ .

$k$  = small value :- wiggly boundary.

$k$  = large value :- smooth boundary.



Bias-variance intuition:

| $k$    | bias     | variance | model behaviour              |
|--------|----------|----------|------------------------------|
| small  | low      | high     | fits to local <u>noise</u> . |
| medium | balanced | balanced | <u>good</u> .                |
| large  | high     | low      | too smooth                   |

KNN parameters,

① K

② distance metric

Distance weighted KNN:

$$w_i = \frac{1}{d(x, x_i) + \epsilon} \quad \epsilon \approx 1e^{-10}$$

$$\hat{y} = \frac{\sum_{i=1}^N w_i y_i}{\sum_{i=1}^N w_i} \rightarrow \text{weighted average.}$$

↓

|       |       |       |       |
|-------|-------|-------|-------|
| $x_1$ | $x_2$ | $x_3$ | $x_4$ |
| 1     | 2     | 3     | 4     |
| 5     | 6     | 7     | 8     |
| → 9   | 10    | ? 11  | 12    |
| 13    | 14    | 15    | 16    |

KNN as imputer.